Short questions

|!|BCESB0090000001|!| ★

For each of the blood vessel type listed in column I, select from column II one phrase that correctly describes it. Put the letter in the space provided. Some may require more than one answer. (5 marks)

Column I		***Column II***
Artery	(a)_____	A. lowest blood pressure
Vein	(b)_____	B. site of exchange of materials between blood and body tissues
Capillary	(c)_____	C. carry blood to organs
		D. with valves
		E. lowest blood flow velocity

##

(a) C (1)

(b) A, D (1, 1)

(c) B, E (1, 1)

(5 marks)

##

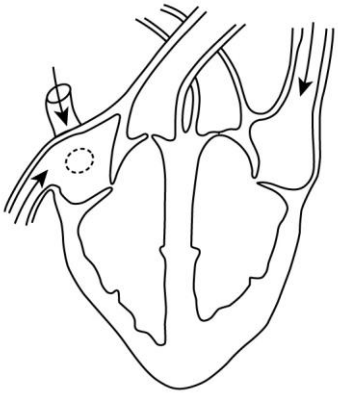
	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

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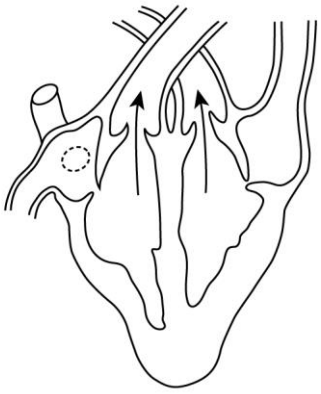
|!|BCESA0090000002|!| ★★

The diagrams below represent the stages in the cardiac cycle. The cardiac cycle is the sequence of events involved from the beginning of one heartbeat to the beginning of the next heartbeat.

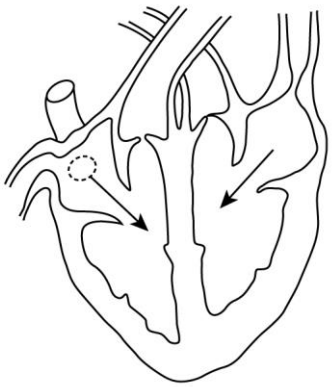
A



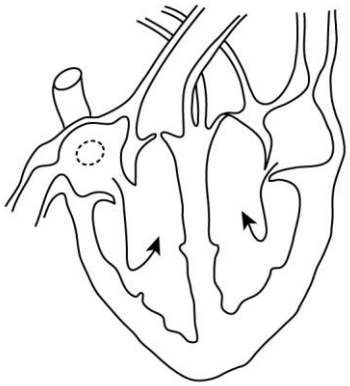
B



C



D



(a) Arrange the diagrams in the correct sequence. (1 mark)

A → _____ → _____ → _____

(b) In which diagrams is the blood pressure increasing in the atria? (2 marks)

_____

(c) What causes the closure of the bicuspid and tricuspid valves in diagram B? (2 marks)

_____

_____

##

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(a) A → C → D → B (1)

(b) A and C (1, 1)

(c) The increased pressure of the blood in the ventricle (1)
caused by the contraction of the ventricles. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Understanding basic concepts	2

|!|BCESB0090000003|!| ★

The photomicrograph below shows an artery and a vein.



(a) Measure the diameter of the artery as indicated in the photomicrograph and calculate the actual diameter of the artery. (2 marks)

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(b) With reference to the photomicrograph, explain how the structure of the artery is related to its function. (2 marks)

(c) State the structure in the vein that is ***not*** shown in the photomicrograph. What is the function of this structure? (2 marks)

##

(a) Diameter of the artery = 28 mm (1)
Actual size = 28 / 20 = 1.4 mm (1)
(b) Its thick wall (1)
helps resist high blood pressure. (1)
(c) valve (1)
to prevent the backflow of blood. (1)

(6 marks)

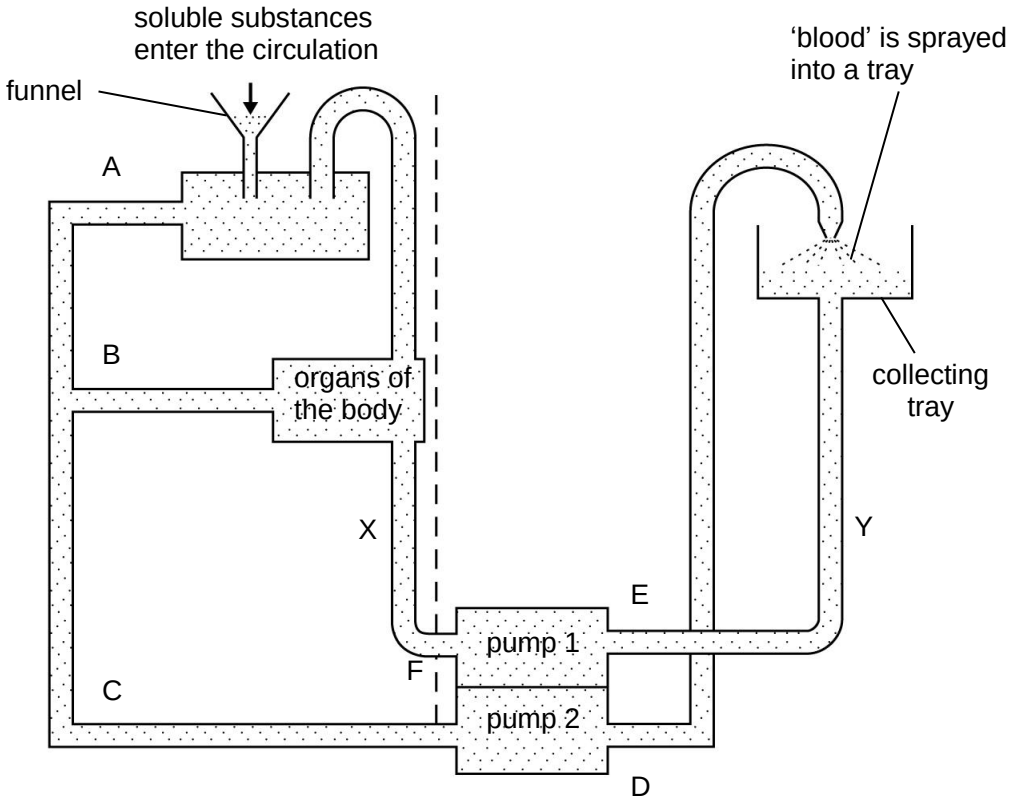
##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Understanding basic concepts	2

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|!|BCESA0090000004|!| ★★

The diagram below shows a model built by a student to represent the pattern of blood circulation in the human body.



(a) In the diagram, draw an arrow at X and another at Y to show the direction of blood flow. (1 mark)

(b) What substance is present in greater amounts in tube E than tube D? Explain your answer. (3 marks)

_____

_____

_____

_____

(c) Which side of the heart does pump 1 represent? (1 mark)

_____

(d) What organ in the human body does the funnel represent? (1 mark)

_____

##

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(a) Correct drawing (1)
- at X: arrow pointing upwards
- at Y: arrow pointing downwards

(b) oxygen (1)
The 'blood' droplets come into contact with air, (1)
and oxygen in the air dissolves in the 'blood'. (1)

(c) left side (1)

(d) small intestine (1)

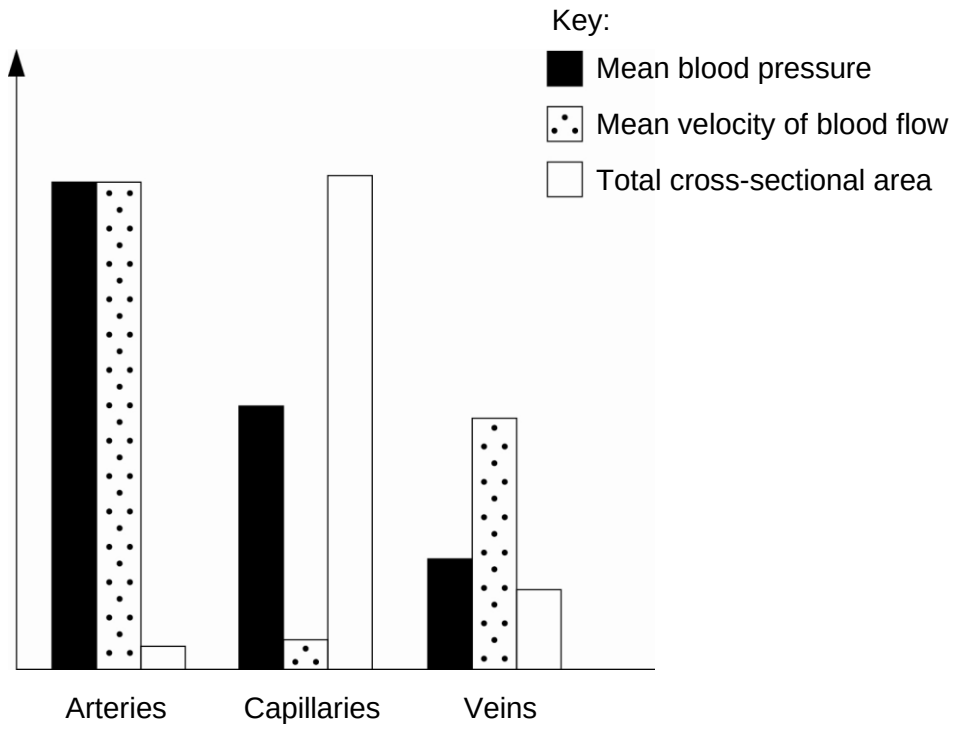
(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	3
(c)	Analysing data	1
(d)	Analysing data	1

|!|BCESA0090000005|!| ★★

The following bar chart shows some information about three kinds of blood vessels.



(a) (i) Which blood vessel has the highest mean blood pressure? (1 mark)

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(ii) State ***one*** feature of this blood vessel that allows it to withstand the high blood pressure. (1 mark)

_____

(b) Explain ***one*** advantage of the total cross-sectional area of the capillaries shown in the graph. (2 marks)

_____

_____

(c) According to the chart above, explain why the mean velocity of blood flow is higher in veins than in capillaries. (2 marks)

_____

_____

##

(a) (i) artery (1)

(ii) It has thick muscular walls. (1)

(b) Any ***one*** set of the following: (2)

1. It provides a greater surface area (1)
   for the diffusion / exchange of materials. (1)
2. The rate of blood flow is decreased. (1)
   This allows more time for exchange of materials. (1)
3. It lowers the blood pressure in the capillary. (1)
   so it will not rupture easily. (1)

(c) Many capillaries connect into fewer veins. (1)
Veins have smaller total cross-sectional area, so the volume of blood flowing through each vein per unit time must be greater. (1)

OR

The reduced cross-sectional area leads to less friction in blood flow, so the mean velocity of blood flow increases. (1)

(6 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Analysing data	2

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|!|BCESC0090000006|!|

Blood is composed of three kinds of blood cells.

(a) Name these ***three*** kinds of blood cells. (3 marks)

(b) State ***one*** function of each kind of blood cells. (3 marks)

##

(a) Red blood cells (1)
White blood cells (1)
Blood platelets (1)

(b) Red blood cells: contain haemoglobin which can carry oxygen from the lungs to all body cells (1)
White blood cells: defence against pathogens (1)
Blood platelets: help in blood clotting at wounds (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	3

|!|BCESB0090000007|!| ★

During long-distance flights, some passengers suffer from swelling of feet.

(a) Explain why this occurs. (3 marks)

(b) Suggest ***one*** prevention measure to this symptom. (1 mark)

##

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(a) Passengers have to sit for a long time / do not get much chance to move their legs. (1)

The lack of skeletal muscle contractions in legs reduces the flow rate of lymph in the lymph vessels. (1)

This leads to the accumulation of excess tissue fluid in the legs and hence the swelling of feet. (1)

(b) Any ***one*** of the following: (1)

- take a short walk every hour (1)
- wear compression socks (1)

(Or other acceptable answers)

(4 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	1

|!|BCESC0090000008|!|

Atherosclerosis is caused by the deposition of fat on the inner wall of the blood vessels. If it occurs in the coronary blood vessels, heart attack may be resulted.

(a) Name ***two*** health problems, other than heart attack, that is associated with atherosclerosis. (2 marks)

(b) If the coronary artery is blocked, the heart will not function properly. Explain briefly. (2 marks)

(c) Name ***three*** adaptations of the blood vessels where exchange of substances takes place. (3 marks)

##

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(a) Any ***two*** of the following: (1 mark each) (2)
High blood pressure / stroke / obesity / diabetes mellitus (Or other reasonable answers)

(b) If the coronary artery is blocked, less oxygen and nutrients are supplied to the cardiac muscle via the coronary artery. (1)
Cardiac muscle thus cannot obtain enough energy to support their actions. (1)

(c) Capillary walls are only one-cell thick. (1)
Capillaries are highly branched. (1)
Capillaries have a very large total cross-sectional area. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	3

|!|BCESC0090000009|!|
Fill in the blanks to complete the following paragraph. (6 marks)

In humans, the blood in the right ventricle is pumped through the (a)_____ to the lungs where gas exchange takes place. (b)_____ in the air sacs diffuses into the blood. Then the blood flows back to the (c)_____ of the heart and leaves the heart again through the (d)_____ to other body organs. In other body organs, the blood becomes (e)_____ and flows back to the (f)_____ of the heart to complete a circulation.

##

(a) pulmonary arteries / artery (1)
(b) Oxygen (1)
(c) left atrium (1)
(d) aorta (1)
(e) deoxygenated (1)
(f) right atrium (1)

(6 marks)

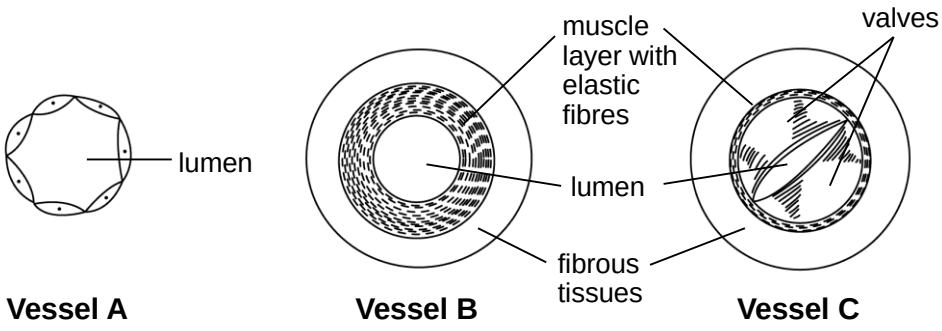
##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESC0090000010|!|

The following diagram shows the transverse sections of three types of blood vessels in humans.



(a) Name vessels A, B and C. (3 marks)

(b) If vessels B and C are blood vessels of a kidney, state ***two*** differences in composition between the blood in vessels B and C. (2 marks)

(c) Describe ***one*** function of the elastic fibres in vessel B. (2 marks)

(d) State ***one*** function of the valves in vessel C. (1 mark)

##

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(a) A: capillary (1)

B: artery (1)

C: vein (1)

(b) Any ***two*** of the following comparisons of other reasonable answers: (1 mark each) (2)

	Vessel B	***Vessel C***
Oxygen content	higher	lower
Carbon dioxide content	lower	higher
Urea content	higher	lower
Content of nutrients	higher	lower

(c) Elastic fibres allow the walls to stretch when the heart contracts and recoil when the heart relaxes. (1)

This helps keep the pressure high to maintain a continuous blood flow. (1)

(d) To prevent the backflow of blood. (1)

(8 marks)

##

	Skills assessed	**Marks**
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	1

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